



Nursing Practice Guideline

Opioid and Ketamine Intravenous Analgesia

Reference #: NPG 8

Scope

Site	Service/Department/Unit	Disciplines
SCGOPHCG	Organisation Wide	Nursing

Introduction

This guideline outlines the procedures and responsibilities of individuals and the health service organisation in relation to Opioid and Ketamine Intravenous Analgesia. It ensures the delivery of a uniform, integrated multidisciplinary approach with management that is consistent with best practice.

Risk Statement

Non-compliance with this guideline will:

Breach legislative requirements	☒	Impact on Patient Quality of Care	☒
Breach National/State/Hospital Policy	☒	Impact on Patient Safety	☒
Breach professional standards	☒	Misconduct	☐
Breach SCGH Mission & Values	☒	Staff Safety	☒
Other:	☐		

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1. General Management

1.1 Legislative Requirements

It is a legal requirement that all disciplines that prescribe or administer medications comply with the Medicines and Poisons Act 2014, Medicines and Poisons Regulations 2016, Pharmacy Act 2010 and Hospital Guidelines. It is a legislative requirement that a valid prescription must exist before Schedule 4 and Schedule 8 medications are administered.

Only valid prescriptions will be utilised by any nursing staff to administer medication except for verbal orders and nurse-initiated medications, as per designated list.

The Medicines and Poisons Regulations Act 2014 requires that:

- A register is maintained so that the amount of each schedule 8 medicine procured, used, supplied or stored is apparent,
- Inventories are undertaken of the Schedule 8 medicines in stock,
- The authorised person (Poisons Permit holder) notifies the Chief Executive Officer in writing if an inventory of stock does not agree with register balance,
- A person who loses a poison (medicine) or from whom a poison is stolen notifies police.

Schedule 4 and schedule 8 medications can be prescribed by a Nurse Practitioner (NP) as relevant to their area of clinical practice and protocol.

Schedule 8 medications must be stored in a locked cupboard and a Registered Nurse, or medication competent Enrolled Nurse is authorised to hold the keys to this cupboard. A Registered Nurse or medication competent Enrolled Nurse is also an authorised person, for the purposes of their employment, to have Schedule 8 medications in their possession, therefore all Schedule 8 drugs ***must be administered by a Registered Nurse, medication competent Enrolled Nurse or registered Medical Practitioner.***

For every transaction, the authorised staff member must record in the register.

- The name of the patient
- The name, quantity and form of the drug,
- The amount discarded if Schedule 8 medications in the form of an injection, vial or infusion are partially used (e.g. Amount issued from stock minus dose administered = amount discarded)
- The date and time of the transaction,
- The register must be ***signed*** and include the ***printed surname*** of each person involved in the transaction,
- The surname of the prescriber,
- The amount of the drug remaining in stock after the transaction (balance)

Alterations, obliterations or cancellations shall NOT be made in the Register of Drugs of Addiction. Any mistake made in any entry should be corrected on the next line available following the entry, signed and printed by two authorised persons and dated.

Where a medicines handling process (e.g. administration) requires two authorised staff members, both staff members are equally responsible and accountable for this documentation.

1.2 Medical Requirements

A valid prescription must be charted by the anaesthetist on the Intravenous Analgesia Chart.

A prescription should only be initiated and altered by:

OPH: an Anaesthetist.

SCGH: A medical member of the Acute Pain Service,

- an Anaesthetist,
- an Intensive Care or Emergency Medicine Practitioner in consultation with an APS medical practitioner.

1.3 Safety Information

Ward-based administration of intravenous opioids is only via Patient Controlled Intravenous Analgesia or Nurse Controlled Analgesia pumps. Large push doses of intravenous opioids are not to be administered on general wards except in an emergency.

SCGH: The Acute Pain Service (APS) is available to review patients Monday to Saturday and as requested by medical and nursing staff ^(1, p.13) After hours, contact the Anaesthesia Registrar.

OPH and SCGH: The on-call Anaesthetist should be contacted to review patients after-hours.

Avoid administration of other opioids by other routes and sedatives whilst the therapy is in progress, unless specifically ordered by the APS, Anaesthetist or Anaesthetic registrar. ^(1, p.13)

Patients who have a known drug dependency (e.g., opioid, alcohol, or benzodiazepine) should be monitored for withdrawal symptoms regardless of the current acute pain management plan.

Naloxone ampoules must be available on the ward when Patient Controlled Intravenous Analgesia (PCA) or Nurse Controlled Analgesia (NCA) is prescribed. ^(1, p.14)

Patient must be escorted by a nurse when away from ward for any reason. If the patient is non-compliant with remaining on the ward the infusion is to be discontinued and:

OPH: Anaesthetist notified

SCGH the Acute Pain Team notified.

1.4 Nursing Management

- The nurse caring for the patient must be proficient in the care and management of patients receiving these therapies. ^(1, p.13)
- Only nurses who have received education and been deemed proficient in the programming and commencement of Patient Controlled Intravenous Analgesia (PCA) may do so, using a valid prescription. ^(1, p.13)
- Use multimodal analgesia i.e., give paracetamol / NSAIDs regularly as prescribed whilst on PCA/NCA, unless contra-indicated.
- Enrolled Nurses that are not medication competent and who have achieved the relevant proficiency may care for patients with PCA / NCA with the exception of:
 - Adjusting the rate of NCA
 - Administering an opioid NCA bolus
 - Programming a PCA pump
 - Changing the opioid or ketamine syringe / bag, and
 - Cessation of NCA / PCA
- Syringes prefilled for bolus use or infusions that are labelled by manufacturer or hospital pharmacy DO NOT require additional labelling.
- The route of administration must be identified / labelled on **all lines**, identifying the date and time that the line was commenced. ^(2, p.22)
- An additives label must be completed and attached IMMEDIATELY to all fluids (bag / burette or syringe) when an injectable medicine is added in the clinical / ward area.

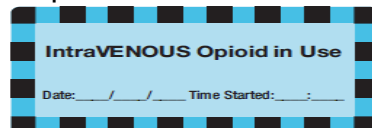
- Information on the additive label must include:
 - Patient name, URN and date of birth
 - The name and quantity of the medication
 - The volume of fluid (mL) total in bag or syringe
 - The type of diluent & concentration (units/mL)
 - Date and time prepared
 - Signature of staff preparing and checking^(2, p. 18)
- The infusion line must have a colour coded Intravenous label and a medicines label attached.

For IntraVENOUS Use Only			
Patient ID		DOB	
Medicine/s	Amount (units)	÷	Volume (mL) = Conc (units/mL)
Diluent			
Date	Prepared by		
Time	Checked by		



DO NOT inject any other drugs through the opioid or ketamine infusion line.

- The patient's medication chart must have a sticker affixed to the PRN section stating 'IntraVENOUS Opioid in Use'.^(1, p. 14)



- When an additional opioid or ketamine infusion is started, observations are to commence from the beginning of the infusion.
- Patients should have oxygen administered, either 5L/min via facemask or 2L/min via nasal prongs, for the duration of opioid therapy or until directed otherwise by APS or anaesthetist.⁽¹⁾ Oxygen therapy may be administered as clinically indicated. Refer to SCGOPHCG Clinical Practice Guideline on Oxygen Therapy and oxygen prescription on patient medication chart.
- Patient's pain should be assessed on both rest and movement and the Functional Activity Score (FAS) should be recorded.
- For inadequate analgesia or any other problems concerned with therapy:
 - OPH contact anaesthetist
 - SCGH contact the Acute Pain Service.
- On cessation, remaining opioid solution is to be discarded by a registered nurse, or medication competent enrolled nurse. This must be witnessed by an authorised person and documented on the Intravenous Analgesia Chart by two staff, one of whom must be a RN or Medical Practitioner.

2. Patient Controlled Intravenous Analgesia (PCA)

2.1 Safety Information

Programming of the PCA pump is the responsibility of APS staff and those who are proficient in the procedure.

SCGH: PCA infusions should be delivered via a locked, dedicated B Braun syringe driver with blue front-plate (available from outside PACU). PCA pump should be set up with syringe and PCA giving set with anti-siphon and anti-reflux valves to prevent drug theft from the syringe.⁽¹⁾

OPH: PCA infusions should be delivered via a locked, dedicated CADD solis pump with a clear giving set with anti-siphon and anti-reflux valves to prevent drug theft from the bag⁽¹⁾.

To ensure patient safety, no one other than the patient should **EVER** operate the PCA handset. If the patient is unable to use PCA effectively, the APS must be contacted promptly to review analgesia. ^(1, p.13)

Programmed values and written order to be checked at the beginning of each shift, after each syringe / bag change and every time that the pump has been opened.⁽¹⁾

Patient must be escorted by a nurse when away from ward for any reason. If the patient is non-compliant with remaining on the ward the infusion is to be discontinued and:

- OPH contact anaesthetist
- SCGH contact the Acute Pain Service.

Nursing Management

- Educate patient prior to starting therapy. Regularly reaffirm patient understanding and monitor use to ensure effective and safe use of PCA.
- The ward nurse will check programmed values and written orders with Post Anaesthetic Care Unit (PACU) Nurse before transferring patient to the ward.
- Program to be checked at beginning of each shift and every time that the pump has been opened. Document 'checked and correct' on the inside of the Intravenous Analgesia Chart each time the program is checked.⁽¹⁾
- PCA syringe / bag is to be changed as required and every 24 hours. PCA tubing to be changed every 72 hours.

OBSERVATIONS

- From commencement of therapy use the Intravenous Analgesia Chart to record:
 - Sedation score
 - Pain at rest and movement
 - Functional activity score (FAS)
 - Nausea score
 - Cumulative drug (measurement in mg/mcg)
- In addition to the regular assessment and recording of core physiological observations, use the Observation Response Chart to record respiratory rate, oxygen saturation and level of consciousness. These observations must be recorded at the following intervals:

Time since commencement	Observation frequency
0-2 hours	Every 30 minutes
2-8 hours	1 hourly
8-24 hours	2 hourly
> 24 hours	As warranted but at least 4 hourly

⁽¹⁾

- Blood pressure and pulse recordings as required post operatively, then 4 hourly if clinically stable.
- After 24 hours, if patient's condition is stable all core physiological observations may be assessed and recorded 4/24.
- Care must be taken to distinguish the over-sedated patient from one in normal sleep. Over-sedation highlights a risk of impending respiratory depression.
- Whenever an observation falls within a shaded area on the Observation Response Chart, the actions required for that colour must be initiated. Respiratory rate 5-9,

consciousness to pain only, and /or if sedation score is greater than 2 requires medical review:

- Remove hand control from patient,
- Attempt to rouse patient,
- Ensure patent airway and administer oxygen 5l/min via Hudson mask (for respiratory rate 5-9) and oxygen 12l/min via Hudson mask (for respiratory rate less than 5), as the current NPGs suggest,
- Prepare Naloxone 400 mcg in 10mls normal saline to be administered as directed.
- If concerned, contact:
 - OPH contact anaesthetist
 - SCGH contact Acute Pain Service or after-hours Anaesthetic Registrar on-call, and Clinical Nurse Consultant.
- If at any time a sedation score of 3 or respirations less than 5:
 - Dial 55 to activate a MET call,
 - Ensure patent airway and administer oxygen 12l/min via Hudson mask,
 - Prepare Naloxone 400 mcg in 10mls normal saline to be administered as directed.

3. Nurse Controlled Analgesia (NCA)

3.1 Safety Information

SCGH: NCA infusions should be delivered via a locked, dedicated B Braun syringe driver with blue front-plate (available from outside PACU). NCA pump should be set up with syringe and PCA/NCA giving set with anti-siphon and anti-reflux valves to prevent drug theft from the syringe.⁽¹⁾

OPH: NCA infusions should be delivered via a locked, dedicated CADD solis pump with a clear giving set with anti-siphon and anti-reflux valves to prevent drug theft from the bag.⁽¹⁾

Program to be checked at beginning of each shift; document 'checked and correct' on the inside of the Intravenous Analgesia Chart each time the program is checked.

Patient must be escorted by a nurse when away from ward for any reason. If the patient is non-compliant with remaining on the ward the infusion is to be discontinued and:

- OPH contact anaesthetist
- SCGH contact the Acute Pain Service.

Nursing Management

NCA infusion fluids are to be changed as required and every 24 hours. Lines must be changed every 72 hours.⁽²⁾

Observations

From the commencement of therapy use the Intravenous Analgesia Chart to record:

- Sedation score,
- Pain at rest and movement
- Functional activity score (FAS)
- Nausea score
- Cumulative drug (measurement in mg/mcg)

In addition to the regular assessment and recording of core physiological observations, use the Observation Response Chart to record respiratory rate, level of consciousness and oxygen saturation. Whenever an observation falls within a shaded area, the actions

required for that colour must be initiated. These observations must be recorded at the following intervals:

Time since commencement	Observation frequency
0-2 hours	Every 30 minutes
2-8 hours	1 hourly
8-24 hours	2 hourly
> 24 hours	As warranted but at least 4 hourly
Post NCA bolus	10 minutes after bolus, hourly respiratory rate and sedation score for 24hrs

Core physiological observations including blood pressure and pulse must be assessed as required post operatively, and then 4 hourly if clinically stable.

Whenever an observation falls within a shaded area on the Observation Response Chart, the actions required for that colour must be initiated. Respiratory rate 5-9, consciousness to pain only and /or if sedation score is greater than 2:

- Cease infusion,
- Attempt to rouse patient,
- Ensure patent airway and administer oxygen 5l/min via Hudson mask,
- If concerned, contact:
 - OPH contact anaesthetist
 - SCGH contact Acute Pain Service or after-hours Anaesthetic Registrar on-call, and Clinical Nurse Consultant
- If at any time a sedation score of 3 or respirations less than 5:
 - Dial 55 to activate a MET call,
 - Ensure patent airway and administer oxygen 12l/min via Hudson mask,
 - Prepare Naloxone 400 mcg in 10mls normal saline to be administered as directed. ⁽²⁾

TITRATION OF ANALGESIA

- If pain is persistent or anticipated, a Registered Nurse or medication competent enrolled nurse may give a bolus dose as charted if sedation score is no more than 2 and respiratory rate at least 8.
- Two nurses (one of whom must be a RN) must remain at the bedside until the bolus dose has been delivered and the opioid infusion has been re-started at the desired rate.
- Assess and record blood pressure, pulse, respiratory rate, pain and sedation scores at 10 minutes after the administration of a bolus dose, with subsequent assessment of respiratory rate and sedation score hourly for 24 hours.
- Doses may be given at 10-minute intervals for 30 minutes. If after this, pain is not controlled, call the APS or the on-call Anaesthetic Registrar.
- Infusion rate should be increased at the time of bolus for persistent pain. Rate changes must be checked by a second nurse and recorded on the Intravenous Analgesia Chart. Titrate the infusion to achieve adequate analgesia.
- NCA infusion rate should be decreased as patient's pain decreases.

4. Ketamine Infusion

4.1 Safety Information

Ketamine may be ordered as a continuous intravenous or subcutaneous infusion by the APS to augment intravenous opioid analgesia. Ketamine antagonises NMDA receptors in

the spinal cord, which when activated, causes an increased sensitivity, and exaggerated response to pain.⁽³⁾ Visual hallucinations may be experienced. Higher doses may result in general anaesthesia.

SCGH: Ketamine is set-up in a locked, single channel B Braun pump with green faceplate, available from outside PACU. Ketamine pumps use a 50ml Terumo syringe and an administration set with anti-siphon and anti-reflux valve, and no Y-connector to prevent drug theft from the syringe.⁽¹⁾

OPH: Ketamine infusions should be delivered via a locked, dedicated CADD solis pump with a clear giving set with anti-siphon and anti-reflux valves to prevent drug theft from the bag⁽¹⁾.

The rate is ONLY to be altered by the APS / Anaesthetist and the patient will not be prescribed a titrated dose.

Programmed values and written order to be checked at the beginning of each shift, after each syringe / bag change and every time that the pump has been opened. Document 'checked and correct' on the inside of the Intravenous Analgesia Chart each time the program is checked.

Patient must be escorted by a nurse when away from ward for any reason. If the patient is non-compliant with remaining on the ward the infusion is to be discontinued and:

- OPH contact anaesthetist
- SCGH contact the Acute Pain Service.

Nursing Management

- Ketamine infusion to be changed as required and every 24 hours. Tubing to be changed every 72 hours.⁽³⁾
- Program to be checked at beginning of each shift and every time that the pump has been opened. Document 'checked and correct' on the inside of the Intravenous Analgesia Chart each time the program is checked.
- Ketamine infusions must be administered using a dedicated line as inadvertent boluses may induce anaesthesia.⁽⁴⁾
- The ketamine infusion line may be connected to a double/ triple lumen peripheral set (i.e. Chook's foot) provided all solutions are compatible.
- Ketamine line must not be attached to the Y-connector on a giving set.
- If intravenous access is not available, administer ketamine via yellow sub-cut butterfly - BD Saf-T-Intima™ with Y Adapter. Refer to NPG #32.
- Ketamine infusions must have an additives label completed and attached.
- Assess pain, functional activity (FAS) and sedation scores, cumulative total hourly for 4 hours then four-hourly. Document on the Intravenous Analgesia Chart (MR810).
- Observe patients for dysphoria, nightmares or hallucinations. If this occurs:
 - OPH contact anaesthetist
 - SCGH contact the Acute Pain Service.
- Oxygen therapy is not required unless clinically indicated

5. Pump Programming Guide (SCGH only)

5.1 Medical Requirements

A valid prescription must be charted using mg/ml and ml/hour by the anaesthetist on the Intravenous Analgesia Chart.

A prescription should only be initiated and altered by:

- A medical member of the Acute Pain Service,
- An Anaesthetist,
- An Intensive care or Emergency medicine practitioner in consultation with an APS medical practitioner.

5.2 Safety Information

Follow all applicable legislation, as well as hospital guidelines for hand hygiene, handling of schedule 8 medications, labelling of intravenous therapy, prevention of healthcare associated infection and confirmation of patient identification.

Ensure that you know the pump code prior to entering the patient area. The pump code **must never** be stated out loud in front of patients.

5.3 Nursing Management

- Only nurses who have received education and been deemed proficient in the programming and commencement of Patient Controlled Intravenous Analgesia (PCA) Nurse Controlled Analgesia (NCA) may do so, using a valid prescription.^(1, p.13)
- Programmed values and written order to be checked at the beginning of each shift, after each syringe change and every time that the pump has been opened. Document 'checked and correct' on the inside of the Intravenous Analgesia Chart (MR 810) each time the program is checked.
- Two nurses, one of whom must be a Registered Nurse must be present at all times when:
 - Programming PCA / Ketamine or NCA infusion,
 - Changing a PCA/ NCA / Ketamine syringe
 - Programming and delivering a PCA or NCA bolus,
 - Ending therapy and switching off pump.

5.4 Procedural Information

Preparing and programming a PCA infusion

Ensure that you know the pump code prior to entering the patient area. The pump code **must never** be stated out loud in front of patients.

1. Collect equipment:
 - Dedicated B Braun syringe driver with blue face plate for PCA infusions (outside PACU)
 - PCA handset
 - 50ml Terumo syringe
 - 50 ml Normal Saline
 - Prescribed opioid
 - PCA/NCA giving set with anti-siphon and anti-reflux valves.⁽¹⁾
 - Line labels and additive label
 - Correct prescription with legible medical practitioner's signature
 - Chooks foot or J-loop if required
2. Prepare and label the opioid syringe as per valid prescription on the Intravenous Analgesia Chart (MR 810).
3. Connect syringe to syringe pump giving set and manually prime the line. Ensure that the giving set has a colour blue coded label designating route of administration and time of commencement.
4. Press on/off button to switch pump on
5. Open front programming window of syringe driver

6. Pull lever out gently and turn to the right
7. Insert syringe with tip facing left and so that the syringe wings sit upright to the left side of the green plastic slot.
8. Replace lever into original position and close front programming window
9. Select Terumo 50ml syringe using left arrow button
- 10. Press up arrow to choose “Yes” at “Choose Drug Library?” prompt**
11. Press left arrow to select “PCA”
12. Select either PCA Fentanyl or PCA Hydromorphone. Pump screen will read “Data lock is active”
13. In the programming screen, program prescribed bolus dose
14. Program prescribed “Lockout Time”. This is always between 6 and 10min.
- 15. No further programming is required**
16. Confirm correct program with 2 nurses, one of whom must be an RN.
17. Lock the Perspex box
18. At patient bedside confirm patient ID by 3 identifiers. 6 “rights” of drug administration to be completed by 2 nurses, one of whom is a registered nurse.
19. Explain procedure for pressing PCA button for analgesia to patient
20. Attach B Braun pump to drip pole and connect giving set to patient
21. Press “start/stop” button to begin infusion

Preparing and Programming a Ketamine Pump

Ensure that you know the pump data code prior to entering the patient area. The pump code **must never** be stated out loud in front of patients.

1. Collect equipment:
 - Dedicated B Braun syringe driver with green faceplate for ketamine infusions (outside PACU)
 - 50ml Terumo syringe
 - 50 ml Normal Saline
 - Ketamine 200mg IV
 - PCA giving set with anti-siphon and anti-reflux valves.⁽¹⁾
 - Line labels and additive label
 - Correct prescription with legible medical practitioner’s signature
 - Chooks foot or J-loop if required
 - Sub-cut butterfly if required
2. Prepare and label the ketamine syringe as per valid prescription on the Intravenous Analgesia Chart.
3. Connect syringe to syringe pump giving set and manually prime the line. Ensure that the giving set has a colour coded label designating route of administration and time of commencement.
4. Press on/off button to switch pump on
5. Open front programming window of syringe driver
6. Pull lever out gently and turn to the right
7. Insert syringe with tip facing left and so that the syringe wings sit to the left of the green plastic slot.
8. Place lever into original position
9. Select Terumo 50ml syringe using arrow button

10. Press up arrow to choose "Yes" at "Choose Drug Library?" prompt
11. Press left arrow to select "Ketamine". Pump screen will read "Data lock is active"
12. In the programming screen use arrow button to select dose if more than 4mg/hr
13. Two nurses (one of which is a RN) independently check and confirm the correct program.
14. Press OK
15. Lock Perspex box
16. At patient bedside two nurses (one of which is a RN) confirm patient ID using 3 identifiers. The 6 "rights" of drug administration must be completed by two (2) nurses. A registered nurse/ medication competent enrolled nurse must handle, attach and commence the infusion.
17. Attach B Braun pump to drip pole and connect giving set to patient
18. Press "start/stop" button to begin infusion

Preparing and programming an NCA infusion

Ensure that you know the pump code prior to entering the patient area. The pump code **must never** to be stated out loud in front of patients.

1. Collect equipment:
 - Dedicated B Braun syringe driver with blue face plate for PCA/NCA infusions (outside PACU)
 - PCA handset should be disconnected and left in PACU area
 - 50ml Terumo syringe
 - 50 ml Normal Saline
 - Prescribed opioid
 - PCA/NCA giving set with anti-siphon and anti-reflux valves.⁽¹⁾
 - Line labels and additive label
 - Correct prescription with legible medical practitioner's signature
 - Chooks foot or J-loop if required
2. Prepare and label the opioid syringe as per valid prescription on the Intravenous Analgesia Chart.
3. Connect syringe to syringe pump giving set and manually prime the line. Ensure that the giving set has a blue colour coded label designating route of administration and time of commencement.
4. Press on/off button to switch pump on
5. Open front programming window of syringe driver
6. Pull lever out gently and turn to the right
7. Insert syringe with tip facing left and so that the syringe wings sit upright in the left-hand side of the green plastic slot and the end of the main pump.
8. Place lever into original position and close front programming window
9. Select Terumo 50ml syringe using left arrow button
- 10. Press up arrow to choose "Yes" at "Choose Drug Library?" prompt**
11. Press down and left arrows to select "NCA"
12. Select either NCA Fentanyl or NCA Hydromorphone. Pump screen will read "Data lock is active"
13. In the programming screen, if bolus volume is correct, scroll down to choose next programming parameter.

14. If bolus volume is not correct, press OK, enter data lock code to change the program and enter prescribed bolus dose. Press OK
15. Check "Lockout Time". This is always 10min.
16. If a background infusion is prescribed, enter the rate per hour by selecting "Dose". Press OK
17. Enter pump code. Then enter the prescribed rate per hour. Press OK.
- 18. No further programming is required**
19. Confirm correct program with 2 nurses, one of whom must be an RN.
20. Lock Perspex box.
21. At patient beside. two nurses (one of which is a RN) confirm patient ID using 3 identifiers. 6 "rights" of drug administration must be completed by 2 nurses. A registered nurse/ medication competent enrolled nurse must handle, attach and commence the infusion.
22. Attach B Braun pump to drip pole and connect giving set to patient
23. Press "start/stop" button to begin infusion

Administering an NCA Bolus

Ensure that you know the pump data code prior to entering the patient area. The pump code **must never** be stated out loud in front of patients.

1. At patient beside. two nurses (one of which is a RN) confirm patient ID using 3 identifiers. Ensure that the 6 "rights" of drug administration are completed by 2 nurses. A registered nurse/ medication competent enrolled nurse may deliver the bolus
2. Check the prescription for amount of opioid to be delivered.
3. Press yellow "Bol" button to access the bolus screen.
4. Enter the data lock code.
5. Using the directional arrow keypad select the amount of bolus to be delivered.
6. Check this amount against the prescription with 2 nursing staff.
7. No further programming is required.
8. Press yellow "Bol" button to commence bolus delivery.
9. The screen will state that the "bolus running: 60ml/hr". Bolus delivery should take 1 min.
10. When bolus is complete the pump will revert to the normal running screen.
11. Document and sign for NCA bolus on front of Intravenous Analgesia Chart.

Changing a PCA/ NCA/ Ketamine syringe

Ensure that you know the pump data code prior to entering the patient area. The pump code **must never** be stated out loud in front of patients.

1. Pump will alarm to say change syringe
2. Prepare and label the opioid syringe as per valid prescription on the Intravenous Analgesia Chart.
3. At patient beside. two nurses (one of whom is a RN) confirm patient ID using 3 identifiers; 6 "rights" of drug administration to be completed by 2 nurses. A registered nurse/ medication competent enrolled nurse must handle, attach and commence the infusion.

4. Unlock the Perspex box.
5. Press stop/start button.
6. Unlock using pump code.
7. Gently pull and release the lever briefly (do not twist) replace back in the same position. This will enable "syringe change prompt".
8. Choose "yes" to syringe change prompt.
9. Syringe driver arm will open and release syringe.
10. Open front programming window: pull out and twist lever gently to release the old syringe.
11. Remove old opioid syringe and insert new syringe, close front programming cover and return lever back into position.
12. Lock the Perspex box
13. Select Terumo syringe
14. Arm will advance and lock syringe in place.
15. Two nurses (one of whom is a RN) independently check and confirm correct program
16. Press start.

6. Cessation of Treatment

6.1 Medical Requirements

Medical officer is to prescribe step down analgesia.

6.2 Nursing Management

PCA / NCA may be ceased by a registered nurse/ medication competent enrolled nurse who is proficient in the procedure and if the patient meets APS cessation criteria.

Cessation Criteria

C = current and planned intervention causes minimal pain.

E = evidence of gut function e.g., tolerating fluids.

A = analgesic requirements are low e.g., PCIA use is low.

S = 'step-down' analgesia is charted.

E = exceptions to nurse cessation are not present i.e., no past or current difficulties with pain management e.g., opioid tolerance, chronic pain, or substance abuse history, need for Ketamine infusion.

- The change of pain management and timing should be discussed with the patient.
- Date, time and signature of registered nurse/ medication competent enrolled nurse ceasing PCA/NCA should be entered in the relevant area of the Intravenous Analgesia Chart.
- Two nurses, one of whom must be a Registered Nurse are required for this procedure.
- The registered nurse and witness (nursing or medical staff) must observe the discarding of partially used opioid solutions, and record volume discarded on the Intravenous Analgesia Chart ^(1, p.15)

Note: OPH the remaining opioid solution is discarded into the red pharmacy medication waste container.

6.3 Procedural Information (SCGH only)

1. Press “Stop” button.
2. Unlock the locked Perspex box.
3. Gently pull and release the lever briefly (do not twist) replace back in the same position. This will enable “syringe change prompt”.
4. Choose “yes” to syringe change prompt.
5. Syringe driver arm will open and release syringe.
6. Open front programming window: pull out and twist lever gently to the right to release the old syringe.
7. Remove old opioid syringe and dispose of remaining drug.
8. Two nurses (one of whom must be a registered nurse) must document and sign for disposal of remaining quantity of schedule 8 medications on front of Intravenous Analgesia chart.
9. Once the old syringe has been removed, return the lever to its original position, then press power button for 3 seconds and the pump will switch off.
10. Infusion pumps should be cleaned using detergent wipes and returned to the Post Anaesthetic Care Unit (PACU).

Related Documents

Legislation:

- Health Practitioner Regulation National Law (WA) Act 2010
- Nursing and Midwifery Board of Australia (NMBA) Code of Professional Conduct for Nurses March 2018
- Western Australia: Medicines and Poisons Act 2014, Medicine and Poisons Regulations 2016, Pharmacy Act 2010
- Work Health and Safety Act 2020

Policy:

- Health Mandatory Policy 0131/20 High Risk Medication Policy.
- NMHS Clinical Documentation Policy
- SCGOPHCG Infection Prevention and Management manual -Aseptic Technique Policy.
- SCGH Patient Identification
- OPH Patient Identification
- NMHS Patient Identification and Procedure Matching Policy
- NMHS Administration of S8 and S4R Medicines Policy and Procedure
- NMHS Destruction, Discard and Disposal of S8 and S4R Medicine Policy
- NMHS Destruction, Discard and Disposal of S8 and S4R Medicine Procedure
- SCGOPHCG Medicines Management Policy and Procedure
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For full details regarding, development, review, consultation, approval, endorsement - contact the Policy Officer for the submission form

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